

[121]

SARDAR PATEL UNIVERSITY

M.Sc.(CA&IT) Examination, 1st Semester

Tuesday, 19th November, 2019.

Time: 10:00 A.M to 01:00 P.M

Subject Code: PS01CIIT32

Subject: Analysis and Design of Algorithm

**Total Marks: 70
[08]**

Q.1 Multiple Choice Questions

1. In Greedy Method the objective function is maximized or minimized the feasible solution is
 - a. Maximized. b. Optimal c. Minimized d. Lower
2. If we do not pick that object to fill into a knapsack then its value is
 - a. Two. b. Zero. c. One. d. None of these.
3. Which of the Following Problem is Dynamic Programming Problem?
 - a. Prime b. Knapsack c. Krushkal d. Kanpsack 0/1
4. We need _____ unit of time to Process each Job.
 - a. One b. Three c. Two d. Four
5. The forward Approach Method is also Known as
 - a. Forward Reasoning. c. Backward Reasoning
 - b. Dynamic Programming d. Optimal Reasoning
6. Leaf node is known as _____ node.
 - a. Terminal. c. Parent.
 - b. Root. d. Non-terminal.
7. A binary tree has at most _____ child.
 - a. Two. c. Three.
 - b. One. d. None of these.
8. In Binary Search Circular Node Known as
 - a. Extrenal Node c. Internal Node
 - b. Square Node d. Round Node

Q.2 Short Questions (Any Seven)

[14]

1. What is stack? Define concept of stack.
2. List Asymptotic Noatation.
3. Write a Divide and Conqure General Algorithm.
4. Define Quicksort ?
5. List Shortest Path Applications.
6. List different algorithm to find spanning tree.
7. Explain Selection of Object in Increasing Order of weight in Kanpsack.
8. List different approach for Multistage Graph.
9. What is Dynamic Programming?

(1)

(P.T.O)

- Q.3 a) Explain Push and POP Operation in stack. [6]
 b) Explain algorithm for insert a value in simple queue and delete a value in simple queue. [6]

OR

- Q.3 b) Explain Pre Order and In order Traversal with Example. [6]

- Q.4 a) Explain Quicksort algorithm in detail. [6]
 b) Write and Explain Binary Search Algorithm in detail. Also check $x=151$, $x=-14$ and $x=9$ are found in the following Entries or Not. [6]
 Entries: -15,-6,0,7,9,23,54,82,101,112,125,131,142,151

OR

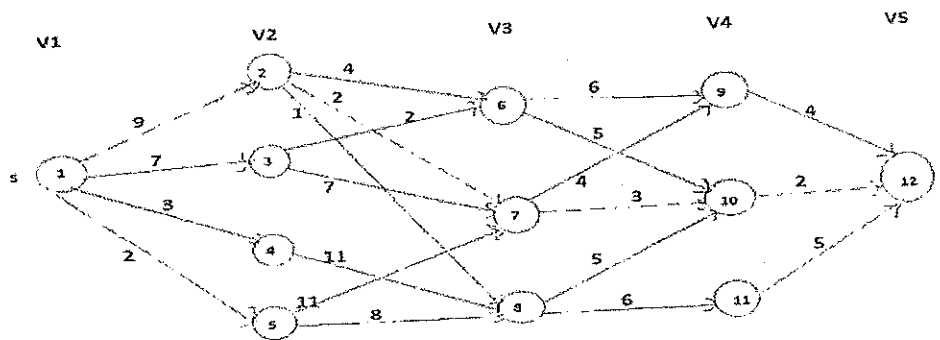
- Q.4 b) Explain Merge sort in detail. [6]

- Q.5 a) Explain General Greedy Algorithm. [6]
 b) Explain Kruskal's Algorithm. [6]

OR

- Q.5 b) Explain Knapsack Algorithm in detail. [6]

- Q.6 a) Solve following example using Forward approach and find shortest path. [6]



MULTI STAGE GRAPH

- b) Explain 0-1 Knapsack Algorithm in detail. [6]

OR

- Q.6 b) Explain Backward Approach in Dynamic Programming. [6]